

APM1220: FA 1 Solutions

Problem 2.2

Given:

$$A = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}$$

(a) $5A$

$$5A = 5 \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} (5)(-1) & (5)(3) \\ (5)(4) & (5)(2) \end{bmatrix} = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}$$

(b) BA

$$BA = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} (4)(-1) + (-3)(4) & (4)(3) + (-3)(2) \\ (1)(-1) + (-2)(4) & (1)(3) + (-2)(2) \\ (-2)(-1) + (0)(4) & (-2)(3) + (0)(2) \end{bmatrix}$$
$$BA = \begin{bmatrix} -4 - 12 & 12 - 6 \\ -1 - 8 & 3 - 4 \\ 2 + 0 & -6 + 0 \end{bmatrix} = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}$$

(c) $A'B'$

First, transpose A and B :

$$A' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix}, \quad B' = \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}$$

Multiply A' by B' :

$$A'B' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix} = \begin{bmatrix} (-1)(4) + (4)(-3) & (-1)(1) + (4)(-2) & (-1)(-2) + (4)(0) \\ (3)(4) + (2)(-3) & (3)(1) + (2)(-2) & (3)(-2) + (2)(0) \end{bmatrix}$$
$$A'B' = \begin{bmatrix} -4 - 12 & -1 - 8 & 2 + 0 \\ 12 - 6 & 3 - 4 & -6 + 0 \end{bmatrix} = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}$$

(d) $C'B$

First, transpose C :

$$C' = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix}$$

Multiply C' by B :

$$C'B = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} (5)(4) + (-4)(1) + (2)(-2) & (5)(-3) + (-4)(-2) + (2)(0) \end{bmatrix}$$
$$C'B = \begin{bmatrix} 20 - 4 - 4 & -15 + 8 + 0 \end{bmatrix} = \begin{bmatrix} 12 & -7 \end{bmatrix}$$

(e) Is AB defined?

No. Matrix A has 2 columns, but matrix B has 3 rows. Matrix multiplication requires the number of columns in the first matrix to equal the number of rows in the second matrix.

Problem 2.4

(a) Prove $(A')^{-1} = (A^{-1})'$

By definition, a matrix multiplied by its inverse equals the identity matrix, I :

$$AA^{-1} = I$$

Take the transpose of both sides:

$$(AA^{-1})' = I'$$

Applying the transpose rule for multiplication, which states $(XY)' = Y'X'$, and since transposing the identity matrix changes nothing ($I' = I$):

$$(A^{-1})'A' = I$$

This equation tells us that when you multiply A' by $(A^{-1})'$, you get the identity matrix. By definition, this means $(A^{-1})'$ is the inverse of A' . Therefore, $(A')^{-1} = (A^{-1})'$.

(b) Prove $(AB)^{-1} = B^{-1}A^{-1}$

To prove this, we multiply (AB) by $(B^{-1}A^{-1})$ to show that the result is the identity matrix I .

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B$$

Since $A^{-1}A = I$:

$$B^{-1}(I)B = B^{-1}B$$

Since $B^{-1}B = I$:

$$(B^{-1}A^{-1})(AB) = I$$

Because their product is the identity matrix, the inverse of (AB) is $(B^{-1}A^{-1})$.

Problem 2.8

Given $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$.

Eigenvalues:

Solve $\det(A - \lambda I) = 0$. First, find λI :

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

Subtract from A :

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix}$$

Find the determinant and solve for λ :

$$(1-\lambda)(-2-\lambda) - (2)(2) = (-2-\lambda+2\lambda+\lambda^2) - 4 = \lambda^2 + \lambda - 6 = 0$$

$$(\lambda+3)(\lambda-2) = 0$$

This gives two eigenvalues: $\lambda_1 = 2$ and $\lambda_2 = -3$.

Eigenvectors:

For $\lambda_1 = 2$, we solve the system $(A - 2I)x = 0$:

$$A - 2I = \begin{bmatrix} 1-2 & 2 \\ 2 & -2-2 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This creates two linear equations:

$$-x_1 + 2x_2 = 0$$

$$2x_1 - 4x_2 = 0$$

From the first equation, we get $x_1 = 2x_2$. If we substitute this into the second equation, we get $2(2x_2) - 4x_2 = 4x_2 - 4x_2 = 0$, confirming x_2 is a free variable.

Let $x_2 = 1$, which makes $x_1 = 2(1) = 2$. The unnormalized eigenvector is $[2, 1]'$.

To normalize it, calculate its length: $\sqrt{2^2 + 1^2} = \sqrt{4 + 1} = \sqrt{5}$. Divide the vector by its length:

$$e_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

For $\lambda_2 = -3$, we solve the system $(A - (-3)I)x = 0$:

$$A + 3I = \begin{bmatrix} 1+3 & 2 \\ 2 & -2+3 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This creates two linear equations:

$$4x_1 + 2x_2 = 0$$

$$2x_1 + x_2 = 0$$

From the second equation, we get $x_2 = -2x_1$. If we substitute this into the first equation, we get $4x_1 + 2(-2x_1) = 4x_1 - 4x_1 = 0$, confirming x_1 is a free variable.

Let $x_1 = 1$, which makes $x_2 = -2(1) = -2$. The unnormalized eigenvector is $[1, -2]'$.

To normalize it, calculate its length: $\sqrt{1^2 + (-2)^2} = \sqrt{1 + 4} = \sqrt{5}$. Divide the vector by its length:

$$e_2 = \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}$$

Spectral Decomposition ($A = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2'$):

$$\begin{aligned} A &= 2 \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} - 3 \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \end{bmatrix} \\ &= 2 \begin{bmatrix} (2/\sqrt{5})(2/\sqrt{5}) & (2/\sqrt{5})(1/\sqrt{5}) \\ (1/\sqrt{5})(2/\sqrt{5}) & (1/\sqrt{5})(1/\sqrt{5}) \end{bmatrix} - 3 \begin{bmatrix} (1/\sqrt{5})(1/\sqrt{5}) & (1/\sqrt{5})(-2/\sqrt{5}) \\ (-2/\sqrt{5})(1/\sqrt{5}) & (-2/\sqrt{5})(-2/\sqrt{5}) \end{bmatrix} \\ &= 2 \begin{bmatrix} 4/5 & 2/5 \\ 2/5 & 1/5 \end{bmatrix} - 3 \begin{bmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{bmatrix} \\ &= 2 \begin{bmatrix} 4/5 & 2/5 \\ 2/5 & 1/5 \end{bmatrix} - 3 \begin{bmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 8/5 & 4/5 \\ 4/5 & 2/5 \end{bmatrix} - \begin{bmatrix} 3/5 & -6/5 \\ -6/5 & 12/5 \end{bmatrix} = \begin{bmatrix} 5/5 & 10/5 \\ 10/5 & -10/5 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} = A \end{aligned}$$

Problem 2.11

Expand the determinant along the first row using cofactors:

$$|A| = a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1p}A_{1p}$$

Since A is a diagonal matrix, all off-diagonal elements in the first row are zero ($a_{12} = 0, \dots, a_{1p} = 0$). This cancels out almost everything, simplifying the formula to:

$$|A| = a_{11}A_{11} + (0)A_{12} + \cdots + (0)A_{1p}$$

$$|A| = a_{11}A_{11}$$

The submatrix A_{11} is also a diagonal matrix. Expanding its determinant in the exact same way yields:

$$A_{11} = a_{22}A_{22}$$

Substituting this back gives:

$$|A| = a_{11}(a_{22}A_{22})$$

Repeating this recursive process for every submatrix down the diagonal results in the product of all main diagonal elements:

$$|A| = a_{11} \cdot a_{22} \cdot a_{33} \cdots a_{pp}$$

Problem 2.24

Given $\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(a) Σ^{-1}

The inverse of a diagonal matrix is formed by taking the reciprocal of each element on the main diagonal:

$$\Sigma^{-1} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/9 & 0 \\ 0 & 0 & 1/1 \end{bmatrix} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Eigenvalues and eigenvectors of Σ

Solve the characteristic equation $\det(\Sigma - \lambda I) = 0$:

$$\lambda I = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\Sigma - \lambda I = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 4 - \lambda & 0 & 0 \\ 0 & 9 - \lambda & 0 \\ 0 & 0 & 1 - \lambda \end{bmatrix}$$

The determinant of a diagonal matrix is the product of its diagonal elements:

$$\det(\Sigma - \lambda I) = (4 - \lambda)(9 - \lambda)(1 - \lambda) = 0$$

Setting each term to zero gives the eigenvalues: $\lambda_1 = 4, \lambda_2 = 9, \lambda_3 = 1$.

To find the eigenvectors, solve $(\Sigma - \lambda I)x = 0$ for each eigenvalue.

For $\lambda_1 = 4$:

$$\begin{bmatrix} 4 - 4 & 0 & 0 \\ 0 & 9 - 4 & 0 \\ 0 & 0 & 1 - 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This yields the equations: $5x_2 = 0 \implies x_2 = 0$ and $-3x_3 = 0 \implies x_3 = 0$. The variable x_1 is free. Let $x_1 = 1$.

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda_2 = 9$:

$$\begin{bmatrix} 4 - 9 & 0 & 0 \\ 0 & 9 - 9 & 0 \\ 0 & 0 & 1 - 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This yields the equations: $-5x_1 = 0 \implies x_1 = 0$ and $-8x_3 = 0 \implies x_3 = 0$. The variable x_2 is free. Let $x_2 = 1$.

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

For $\lambda_3 = 1$:

$$\begin{bmatrix} 4 - 1 & 0 & 0 \\ 0 & 9 - 1 & 0 \\ 0 & 0 & 1 - 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This yields the equations: $3x_1 = 0 \implies x_1 = 0$ and $8x_2 = 0 \implies x_2 = 0$. The variable x_3 is free. Let $x_3 = 1$.

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(c) Eigenvalues and eigenvectors of Σ^{-1}

Solve $\det(\Sigma^{-1} - \lambda I) = 0$:

$$\Sigma^{-1} - \lambda I = \begin{bmatrix} 1/4 - \lambda & 0 & 0 \\ 0 & 1/9 - \lambda & 0 \\ 0 & 0 & 1 - \lambda \end{bmatrix}$$

$$\det(\Sigma^{-1} - \lambda I) = \left(\frac{1}{4} - \lambda\right) \left(\frac{1}{9} - \lambda\right) (1 - \lambda) = 0$$

Setting each term to zero gives the eigenvalues: $\lambda_1 = 1/4, \lambda_2 = 1/9, \lambda_3 = 1$.

To find the eigenvectors, solve $(\Sigma^{-1} - \lambda I)x = 0$ for each eigenvalue.

For $\lambda_1 = 1/4$:

$$\begin{bmatrix} 1/4 - 1/4 & 0 & 0 \\ 0 & 1/9 - 1/4 & 0 \\ 0 & 0 & 1 - 1/4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -5/36 & 0 \\ 0 & 0 & 3/4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This yields the equations: $(-5/36)x_2 = 0 \implies x_2 = 0$ and $(3/4)x_3 = 0 \implies x_3 = 0$. The variable x_1 is free. Let $x_1 = 1$.

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda_2 = 1/9$:

$$\begin{bmatrix} 1/4 - 1/9 & 0 & 0 \\ 0 & 1/9 - 1/9 & 0 \\ 0 & 0 & 1 - 1/9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5/36 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 8/9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This yields the equations: $(5/36)x_1 = 0 \implies x_1 = 0$ and $(8/9)x_3 = 0 \implies x_3 = 0$. The variable x_2 is free. Let $x_2 = 1$.

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

For $\lambda_3 = 1$:

$$\begin{bmatrix} 1/4 - 1 & 0 & 0 \\ 0 & 1/9 - 1 & 0 \\ 0 & 0 & 1 - 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3/4 & 0 & 0 \\ 0 & -8/9 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This yields the equations: $(-3/4)x_1 = 0 \implies x_1 = 0$ and $(-8/9)x_2 = 0 \implies x_2 = 0$. The variable x_3 is free. Let $x_3 = 1$.

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Problem 2.30

Given:

$$\mu_X = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}, \quad \Sigma_X = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix}$$

Partitions: $X^{(1)} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$, $X^{(2)} = \begin{bmatrix} X_3 \\ X_4 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$

(a) $E(X^{(1)})$

$$E(X^{(1)}) = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

(b) $E(AX^{(1)})$

$$E(AX^{(1)}) = AE(X^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = (1)(4) + (2)(3) = 4 + 6 = 10$$

(c) $Cov(X^{(1)})$ We extract the variables associated with X_1 and X_2 from the top-left 2×2 partition of Σ_X :

$$Cov(X^{(1)}) = \Sigma_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

(d) $Cov(AX^{(1)})$

$$Cov(AX^{(1)}) = A\Sigma_{11}A' = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

First multiply A and Σ_{11} :

$$A\Sigma_{11} = [(1)(3) + (2)(0) \quad (1)(0) + (2)(1)] = \begin{bmatrix} 3 & 2 \end{bmatrix}$$

Next multiply by A' :

$$(A\Sigma_{11})A' = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = (3)(1) + (2)(2) = 3 + 4 = 7$$

(e) $E(X^{(2)})$

$$E(X^{(2)}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(f) $E(BX^{(2)})$

$$E(BX^{(2)}) = BE(X^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} (1)(2) + (-2)(1) \\ (2)(2) + (-1)(1) \end{bmatrix} = \begin{bmatrix} 2 - 2 \\ 4 - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

(g) $Cov(X^{(2)})$ We extract the variables associated with X_3 and X_4 from the bottom-right 2×2 partition of Σ_X :

$$Cov(X^{(2)}) = \Sigma_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$$

(h) $Cov(BX^{(2)})$

$$Cov(BX^{(2)}) = B\Sigma_{22}B'$$

First multiply B and Σ_{22} :

$$B\Sigma_{22} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} (1)(9) + (-2)(-2) & (1)(-2) + (-2)(4) \\ (2)(9) + (-1)(-2) & (2)(-2) + (-1)(4) \end{bmatrix} = \begin{bmatrix} 9 + 4 & -2 - 8 \\ 18 + 2 & -4 - 4 \end{bmatrix} = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix}$$

Next multiply by B' :

$$(B\Sigma_{22})B' = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} (13)(1) + (-10)(-2) & (13)(2) + (-10)(-1) \\ (20)(1) + (-8)(-2) & (20)(2) + (-8)(-1) \end{bmatrix}$$

$$Cov(BX^{(2)}) = \begin{bmatrix} 13 + 20 & 26 + 10 \\ 20 + 16 & 40 + 8 \end{bmatrix} = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}$$

(i) $Cov(X^{(1)}, X^{(2)})$ We extract the variables crossing X_1, X_2 with X_3, X_4 from the top-right 2×2 partition of Σ_X :

$$Cov(X^{(1)}, X^{(2)}) = \Sigma_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$$

(j) $Cov(AX^{(1)}, BX^{(2)})$

$$Cov(AX^{(1)}, BX^{(2)}) = A\Sigma_{12}B'$$

First multiply A and Σ_{12} :

$$A\Sigma_{12} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} = [(1)(2) + (2)(1) \quad (1)(2) + (2)(0)] = [2 + 2 \quad 2 + 0] = [4 \quad 2]$$

Next multiply by B' :

$$(A\Sigma_{12})B' = [4 \quad 2] \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = [(4)(1) + (2)(-2) \quad (4)(2) + (2)(-1)]$$

$$Cov(AX^{(1)}, BX^{(2)}) = [4 - 4 \quad 8 - 2] = [0 \quad 6]$$

Problem 2.41

Given:

$$\mu_X = \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix}, \quad \Sigma_X = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} = 3I, \quad A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}$$

(a) $E(AX)$

$$E(AX) = A\mu_X = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} (1)(3) + (-1)(2) + (0)(-2) + (0)(0) \\ (1)(3) + (1)(2) + (-2)(-2) + (0)(0) \\ (1)(3) + (1)(2) + (1)(-2) + (-3)(0) \end{bmatrix}$$

$$E(AX) = \begin{bmatrix} 3 - 2 + 0 + 0 \\ 3 + 2 + 4 + 0 \\ 3 + 2 - 2 + 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}$$

(b) $Cov(AX)$

$$Cov(AX) = A\Sigma_X A' = A(3I)A' = 3(AA')$$

First compute AA' :

$$AA' = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(1) + (-1)(-1) + (0)(0) + (0)(0) & (1)(1) + (-1)(1) + (0)(-2) + (0)(0) & (1)(1) + (-1)(1) + (0)(1) + (0)(-3) \\ (1)(1) + (1)(-1) + (-2)(0) + (0)(0) & (1)(1) + (1)(1) + (-2)(-2) + (0)(0) & (1)(1) + (1)(1) + (-2)(1) + (0)(-3) \\ (1)(1) + (1)(-1) + (1)(0) + (-3)(0) & (1)(1) + (1)(1) + (1)(-2) + (-3)(0) & (1)(1) + (1)(1) + (1)(1) + (-3)(-3) \end{bmatrix}$$

$$= \begin{bmatrix} 1 + 1 + 0 + 0 & 1 - 1 + 0 + 0 & 1 - 1 + 0 + 0 \\ 1 - 1 + 0 + 0 & 1 + 1 + 4 + 0 & 1 + 1 - 2 + 0 \\ 1 - 1 + 0 + 0 & 1 + 1 - 2 + 0 & 1 + 1 + 1 + 9 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

Next, multiply the entire matrix by the scalar 3:

$$Cov(AX) = 3 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix} = \begin{bmatrix} (3)(2) & (3)(0) & (3)(0) \\ (3)(0) & (3)(6) & (3)(0) \\ (3)(0) & (3)(0) & (3)(12) \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

(c) Which pairs of linear combinations have zero covariances?

The covariance between any two different linear combinations is found in the off-diagonal elements of the $Cov(AX)$ matrix we just calculated. Because every off-diagonal element is exactly 0, we can conclude that all pairs of linear combinations have zero covariances.